

Topic 07 Algebra Linear

2)

$$A = \begin{bmatrix} -2 & 3 & 1 & -1 \\ 0 & 1 & 2 & 3 \\ 1 & -1 & 1 & -2 \\ 0 & 3 & 7 & -1 \end{bmatrix}$$

$$L_4 \leftarrow [-2]L_1 + L_4$$

$$\begin{bmatrix} -2 & 3 & 1 & -1 \\ 0 & 1 & 2 & 3 \\ 1 & -1 & 1 & -2 \\ 0 & 3 & 7 & -1 \end{bmatrix}$$

$$L_3 \leftarrow [-1]L_2 + L_3$$

$$\begin{bmatrix} 1 & -1 & 1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & -8 \\ 0 & 0 & 1 & -10 \end{bmatrix}$$

$$L_4 \leftarrow [-3]L_2 + L_4$$

$$\begin{bmatrix} -2 & 3 & 1 & -1 \\ 0 & 1 & 2 & 3 \\ 1 & -1 & 1 & -2 \\ 0 & 0 & 1 & -10 \end{bmatrix}$$

$$L_4 \leftarrow [-1]L_3 + L_4$$

$$\begin{bmatrix} 1 & -1 & 1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & -8 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 & -2 \\ 0 & 1 & 2 & 3 \\ -2 & 3 & 1 & -1 \\ 0 & 0 & 1 & -10 \end{bmatrix}$$

$$\det(A) = 1 \cdot (-2) = -2$$

$$L_3 \leftarrow [2]L_1 + L_3$$

$$\begin{bmatrix} 1 & -1 & 1 & -2 \\ 0 & -1 & 2 & 3 \\ 0 & 1 & 9 & -5 \\ 0 & 0 & 1 & -10 \end{bmatrix}$$



5) B)

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$$B = \left[\begin{array}{ccc|ccc} -3 & 4 & -5 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 3 & -5 & 4 & 0 & 0 & 1 \end{array} \right]$$

$L_3 \leftarrow L_1 + L_3$

$$\left[\begin{array}{ccc|ccc} -3 & 4 & -5 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & -1 & -1 & 1 & 0 & 1 \end{array} \right]$$

$L_3 \leftarrow L_2 + L_3$

$$\left[\begin{array}{ccc|ccc} -3 & 4 & -5 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right]$$

$L_1 \leftarrow [-4] L_2 + L_1$

$$\left[\begin{array}{ccc|ccc} -3 & 0 & -13 & 1 & -4 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right]$$

$L_1 \leftarrow [13] L_3 + L_1$

$$\left[\begin{array}{ccc|ccc} -3 & 0 & 0 & 14 & -9 & 13 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right]$$

$L_2 \leftarrow L_3 + L_2$

$$\left[\begin{array}{ccc|ccc} -3 & 0 & 0 & 14 & -9 & 13 \\ 0 & 1 & 0 & -2 & -1 & -2 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{14}{3} & 3 & -\frac{13}{3} \\ 0 & 1 & 0 & -2 & -1 & -2 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right]$$